

Shunsuke Kikuchi

Tokyo, Japan | kikushun2111@gmail.com | [LinkedIn](#) | [GitHub](#) | [Personal Website](#)

Education

Imperial College London

Medical Robotics & Image-Guided Intervention MRes.

University of California, Los Angeles

B.S in Computational and Systems Biology & B.S in Applied Mathematics (Double Major)

Graduated With College Honors

Arkansas State University

B.S. in Exercise Science

London, United Kingdom

Expected Graduation: September 2027

Los Angeles, CA, United States

Graduation: June 2025

Jonesboro, AR, United States

(Left for transfer, August 2023)

Publication

- **Kikuchi, S.**, Kouno, A., & Matsuzaki, H. "Stitch-Inferencer: Enhance Endoscopic Video Segmentation and Tracking via Panoramic Reconstruction." *In MICCAI Workshop SafeSurg*, 2026.
- **Kikuchi, S.** "Solution for MVAA Challenge 2026: Ensemble, Tool-paste Augmentation and Tool-free View Generation." *In MICCAI Workshop Medical World Models*, 2026.
- Cheng, M., Tran, J., Littman, R., **Kikuchi, S.**, Ha, S. M., Yang, S., et al. "scGRNdb: A Database for over 1300 Cell-type-specific Global Gene Regulatory Networks across Human and Mouse Tissues Profiled in Single Cell Atlases." Under review in *Cell Systems*, 2026.
- **Kikuchi, S.**, Kouno, A., & Matsuzaki, H. "Cholec80-port: A Geometrically Consistent Trocar Port Segmentation Dataset for Robust Surgical Scene Understanding." To appear in IJCARS, Supplement: Proceedings of CARS 2026.
- Rueckert, T., Rauber, D., Maerkl, R., Klausmann, L., Yildiran, S. R., **Kikuchi, S.**, et al. "Comparative validation of surgical phase recognition, instrument keypoint estimation, and instrument instance segmentation in endoscopy: Results of the PhaKIR 2024 challenge." *Medical Image Analysis*, 103945, 2026.
- Yang, K., Shi, P., Huang, H., Musio, F., Baazaoui, H., Aydin, O. U., ..., **Kikuchi, S.**, et al. "TopBrain Segmentation Challenge for Whole Brain Vessel Anatomy." *medRxiv*, 2026.05.28.26354312, 2026.
- Hoffmann, H., Bady, S., de Boer, C., Kirchner, M., Egger, J., ..., **Kikuchi, S.**, et al. "OSS: Open Suturing Skills Vision-Based Assessment Challenge 2024-2025." *arXiv preprint arXiv:2605.22200*, 2026.
- Schmidt, A., Karaoglu, M. A., Sinha, S., Jang, M., Ha, H., Jung, K., Gu, K., **Kikuchi, S.**, et al. "Point Tracking in Surgery: The 2024 Surgical Tattoos in Infrared (STIR) Challenge." *arXiv preprint arXiv:2503.24306*, 2025.
- **Kikuchi, S.**, Kouno, A., & Matsuzaki, H. "Memory-Enhanced Temporal Learning: Leveraging SAM2's Memory Modules for Consistent Video Segmentation." *In MICCAI Workshop MSB EMERGE*, 2025.
- Ju, X., Daher, R., Caramalau, R., Huang, B., Ghamsarian, N., **Kikuchi, S.**, et al. "SegCol Challenge: Semantic Segmentation for Tools and Fold Edges in Colonoscopy data." *arXiv preprint arXiv:2412.16078*, 2024.

Research Experience

Jmees Inc., Tokyo, Japan

Research Engineer, Oct 2025 - Present,

Responsible for R&D to integrate the knowledge gained from our research and the MICCAI challenge into the company's product AI model.

Developed a video segmentation model utilizing image stitching for an expanded field of view to generate tool-free view, enhance surgical video understanding and consistent segmentation in endoscopic surgery.

Research Internship, June 2024 - Sep 2025,

Developed video segmentation model for ureter segmentation for hysterectomy Endoscopic surgery support system, affiliated with National Cancer Center Japan.

Participated in the MICCAI Challenge on surgical video understanding and segmentation themes, accumulating knowledge, and simultaneously promoting the company to the research community.

National Cancer Center Japan Hospital East, Gynecology Department, Tokyo, Japan

Project Researcher under Dr. Shin Takesita (part-time), April 2025 - Present

Conducting research on surgical navigation with clinical researchers and surgery experts, focusing on phase classification-based surgical skill evaluation in Total Hysterectomy and Pelvic Lymphadenectomy.

Institute for Quantitative and Computational Biosciences, University of California, Los Angeles

Undergraduate Researcher under Dr. Xia Yang, October 2024 - June 2025

scGRNdb Project: Constructed spatial transcriptomics GRN analysis pipeline and validation (scanpy, squidpy). Developed interactive web server using PHP, JavaScript, Neo4j, Docker, and Python.

Bruins-In-Genomics (B.I.G.) Summer Program, URC-sciences Summer Scholar, 2024 Summer

Developed and optimized a Python/Shell-based pipeline for gene regulatory network (GRN) prediction using scGRNdb and disease modeling, benchmarking performance with perturb-seq data and adapting resolution-based community detection.

Arkansas Biosciences Institute / NYITCOM / Arkansas State University tri-institutional Research Labs

Undergraduate Researcher under Dr. Jennifer Xie, October 2021- August 2023

Mouse Project: Studied RgIA5474 on Sciatic Nerve Injury and RgIA4 on Cephalic pain, to uncover mechanism of pain in Central Nervous Systems. Performed wet experiments including injection, surgery, blood collection, ELISA, and cardiac perfusion. Responsible for planning experiments schedule, managing materials and colony as a **project leader** in June to August 2023.

Conferences/ Research Presentation

- “Stitch-Inferencer: Enhance Endoscopic Video Segmentation and Tracking via Panoramic Reconstruction”
 - MICCAI 2026 Workshop SafeSurg., Oct., 2026, Strasbourg, France. **Poster Presentation**
- “Solution for MVAA Challenge 2026: Ensemble, Tool-paste Augmentation and Tool-free View generation”
 - MICCAI 2026 Workshop on Medical World Models., Sep., 2026, Strasbourg, France. **Poster Presentation**
- “Cholec80-port: A Geometrically Consistent Trocar Port Segmentation Dataset for Robust Surgical Scene Understanding”
 - Computer Assisted Radiology and Surgery, 2026, Jul., 2026, Nagoya, Japan. **Oral Presentation**
- “Memory-Enhanced Temporal Learning: Leveraging SAM2’s Memory Modules for Consistent Video Segmentation”
 - The 1st Japan-US Science Forum in Southern California, Feb. 2025, UCLA. **Poster Presentation**
 - MICCAI 2025 workshop MSB EMERGE, Oct. 2025, Daejeon, Korea. **Poster Presentation.**
- ” scGRNdb - Cell Type Level Gene Regulatory Network Database for Single-cell Analysis Framework”
 - SPUR Summer Research Showcase 2024, Aug. 2024, UCLA. **Oral Presentation.**
 - BIG Summer Final Session, Aug. 2024, UCLA. **Poster Presentation.**
- “Potential effects of $\alpha 9\alpha 10$ nicotinic receptor antagonists on chronic pain and motor deficits in mice with sciatic nerve injury.”
 - Create @ State 2024 Symposium, Arkansas State University, Apr 2024, **Poster Presentation**
- “Inhibition of cephalic pain by RgIA4, a novel, selective $\alpha 9\alpha 10$ nicotinic acetylcholine receptor (nAChR) antagonist”
 - Create @ State 2023 Symposium, Arkansas State University, Apr 2023, **Poster Presentation**
 - ABI Undergraduate Research Scholar Program Final Session, May 2022, **Oral Presentation**
- Top-Solution Presentation (**Oral Presentation**)
 - MICCAI 2026 Satellite Event “Orena FOCUS challenge”, Sep., 2026, Strasbourg, France
 - MICCAI 2026 Satellite Event “MultiBypass Surgical Action Triplet Challenge”, Sep., 2026, Strasbourg, France
 - MICCAI 2026 Satellite Event “Endoscopic Vision Challenge Cluster 2026 - CLiMB Challenge”, Sep., 2026, Strasbourg, France
 - MICCAI 2025 Satellite Event “SAGES Critical View of Safety Challenge”, Sep., 2025, Daejeon, Korea
 - MICCAI 2025 Satellite Event “TopBrain Segmentation Challenge for Whole Brain Vessel Anatomy”, Sep., 2025, Daejeon, Korea
 - MICCAI 2025 Satellite Event “Endoscopic Vision Challenge Cluster 2025”, Sep., 2025, Daejeon, Korea (4 tasks from OSS and RARE)
 - MICCAI 2024 Satellite Event “Endoscopic Vision Challenge Cluster 2024”, Oct., 2024, Marrakech, Morocco (8 tasks from SegCol, OSS, STIR and PhaKIR)

Activity & Achievement

Kaggle Competition: Expert, Highest Rank: **2,552/205,373**

Leash Bio - Predict New Medicines with BELKA (Silver medal, 27/1,950, 2024 Apr. – Jul.)

HMS - Harmful Brain Activity Classification (Bronze medal, 245/2,767, 2024 Jan. – Apr.)

Conference Challenges:

MICCAI 2026, Endoscopic Vision Challenge Cluster (Sep. 2026)

CLiMB Challenge (2nd)

MICCAI 2026, **Orena FOCUS Challenge** (Frame: **4th**, PROCEDURE: **4th**), Sep. 2026

MICCAI 2025, **SAGES CVS Challenge** (TaskA (GPU): **2nd**, TaskB (CPU): **3rd**), Sep. 2025

MICCAI 2025, Endoscopic Vision Challenge Cluster, Sep. 2025

Open Suturing Skills Challenge (OSS25) (Task1 (GRS): **2nd**, Task2 (OSATS): **1st**, Task3 (Keypoint): **1st**,)

Recognition of Abnormalities in low-pREvalence cancer (RARE25) (2nd)

MICCAI 2025, **TopBrain Challenge** (3rd), Sep. 2025

MICCAI 2024, Endoscopic Vision Challenge Cluster (Oct. 2024)

Semantic Segmentation for Tools and Fold Edges in Colonoscopy data (Segmentation: **1st**, Active Learning: **2nd**)

PhaKIR - Phase, Keypoint and Instrument Recognition (Phase: **2nd**, Instrument: **1st**)

Open Suturing Skills Challenge (OSS) (Task1 (GRS): **3rd**, Task2 (OSATS): **3rd**)

Surgical Tissue Tracking Using the STIR (Surgical Tattoos in Infrared) Dataset (2D: **3rd**, 3D: **2nd**)

6th National Medical AI Contest in Japan (**4th**, Mar. 2024)